

Background.

Let f be analytic on $B_r(z_0) \setminus \{z_0\}$. Then, there is the Laurent's Expansion:

$$f(z) = \sum_{n=-\infty}^{\infty} a_n \cdot (z-z_0)^n \quad (0 < |z-z_0| < r)$$

where $a_n = \frac{1}{2\pi i} \int_C \frac{f(z)}{(z-z_0)^{n+1}} dz$ ($n \in \mathbb{Z}$), C is any closed contour around z_0 .

So that $a_{-1} = \frac{1}{2\pi i} \int_C f(z) dz$. Therefore, if we can find the Laurent's expansion, then we can evaluate integral.
 $\stackrel{\text{def}}{=} \text{Res}(f; z_0)$

Thm. Suppose that f is analytic inside and on a simple closed contour C , except for isolated singularities at $z_1, \dots, z_n$ inside C . Then

$$\int_C f(z) dz = 2\pi i \sum_{k=1}^n \text{Res}(f, z_k).$$

There are two useful Lemmas.

Lemma. If f has a pole of order m at z_0 , then

$$\text{Res}(f, z_0) = \frac{1}{(m-1)!} \cdot \lim_{z \rightarrow z_0} \left(\frac{d}{dz} \right)^{m-1} (z-z_0)^m f(z).$$

Proof. Observe that $f(z) = \frac{a_{-m}}{(z-z_0)^m} + \dots + \frac{a_{-1}}{z-z_0} + a_0 + a_1(z-z_0) + \dots$ ($a_{-m} \neq 0$)

so that $(z-z_0)^m \cdot f(z) = a_{-m} + a_{-m+1}(z-z_0) + \dots + a_{-1}(z-z_0)^{m-1} + g(z) \cdot (z-z_0)^m$, where g is analytic at z_0 .

Now, differentiating $m-1$ times, and evaluating z_0 . $\square$

Lemma. Let f and g be analytic at z_0 .

If g has a simple zero at z_0 and $f(z_0) \neq 0$, then

$$\text{Res}(f/g, z_0) = \frac{f(z_0)}{g'(z_0)}$$

Proof. Since g has simple zero at z_0 , $1/g$ has a simple pole at z_0 . $\square$

